Register No.							

CHRIST (DEEMED TO BE UNIVERSITY), BANGALORE
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
TEST-1 EXAMINATION – JUNE 2026
M.TECH – III SEMESTER

Programme: M. Tech in CSE
Course Name: Advanced Data Mining **Max. Marks: 50**
Course Code: MTCS344E01 **Time: 2hrs**

General Instructions

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / Course title & number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place in the hall.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer booklet. No sheets should be detached from the answer booklet.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer booklet.

Course Outcomes (COs): The students will be able to
CO1: Explain the basic fundamentals of Data mining and how to preprocess the data. (L2)
CO2: Apply appropriate association algorithm to mine frequent patterns. (L3)
CO3: Apply classification and related data mining techniques accurately and efficiently using Python for computations. (L3)
CO4: Apply clustering and related data mining techniques accurately and efficiently using Python for computations. (L3)
CO5: Develop a model using appropriate algorithm to solve real world problems. (L3)

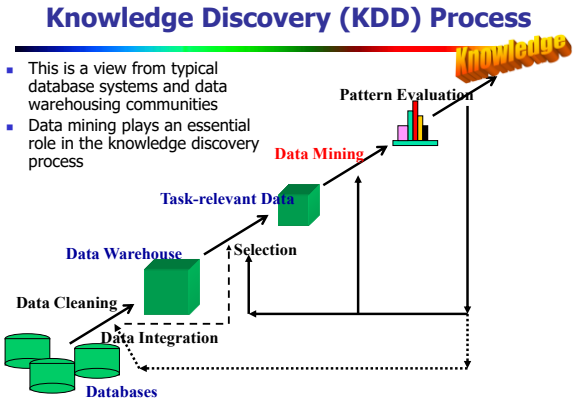
SECTION A

Answer ALL the questions **4 X 10 marks = 40 Marks**

Q. No	Questions	CO	RBT
1	Demonstrate the steps involved in Data Mining when viewed as a process of knowledge discovery with a neat diagram. **In Data science, following are the steps:** ■ **1. *Gathering data*-** The first step in the process is to gather data. It can be structured, unstructured or semi-structured. ■ **2. *Data munging*-** Once you've got your data, it's time to work on it. The 'raw' data is cleaned and transformed into an understandable format to get the most value out of it. This is probably the longest step. Data scientists report data cleaning is about 80% of the time spent on the whole process.	1	L2

3. *Analyzing data*- After cleaning the data, now is the time to analyze it by applying algorithms and necessary statistical models.

- **4. *Data visualization*-** When large amounts of data are to be dealt with, building visualizations or graphs are the best way to explore and communicate results.
- **5. *Producing predictions*-** Machine learning algorithms help you getting insights and predicting future trends. More than just creating predictions, this step can help you build new products and processes.
- **6. *Recapitulate*-** Insights help to develop more features so as to continuously improve model outputs and deliver timely performance and accurate results.



11

2

Discover the correlation coefficient (Pearson's product moment coefficient) from the following table. Are these two attributes positively or negatively correlated? Compute their covariance.

Subject	AGE X	GLUCOSE LEVEL Y
1	43	99
2	21	65
3	25	79
4	42	75
5	57	87
6	59	81

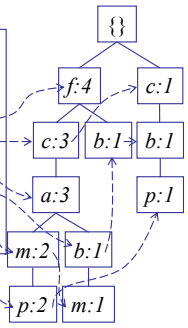
Step 1: Given Data

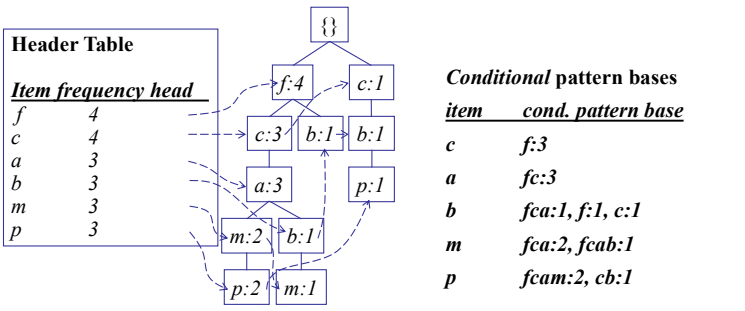
Subject	Age (X)	Glucose Level (Y)	X^2	Y^2	XY
1	43	99	1849	9801	4257
2	21	65	441	4225	1365
3	25	79	625	6241	1975
4	42	75	1764	5625	3150
5	57	87	3249	7569	4959
6	59	81	3481	6561	4779
Total	**247**	**486**	**11409**	**40022**	**20485**

Step 2: Pearson Correlation Formula

$$r = \frac{n\sum XY - \sum X \sum Y}{\sqrt{(n\sum X^2 - (\sum X)^2)(n\sum Y^2 - (\sum Y)^2)}}$$

1 | L2

	Where: • $n = 6$ • $\sum X = 247$ • $\sum Y = 486$ • $\sum X^2 = 11409$ • $\sum Y^2 = 40022$ • $\sum XY = 20485$ $\boxed{r \approx 0.530}$		
3	A database has five transactions as shown below. Let min_sup count = 3 and min_conf = 80%. ***TID*** ***Items bought*** **100** ***{f, a, c, d, g, i, m, p}*** **200** ***{a, b, c, f, l, m, o}*** **300** ***{b, f, h, j, o, w}*** **400** ***{b, c, k, s, p}*** **500** ***{a, f, c, e, l, p, m, n}*** Determine all frequent itemsets and strong association rule for the above dataset using FP-Growth algorithm. **Construct FP-tree from a Transaction Database** *TID* *Items bought* *(ordered) frequent items* 100 {f, a, c, d, g, i, m, p} {f, c, a, m, p} 200 {a, b, c, f, l, m, o} {f, c, a, b, m} 300 {b, f, h, j, o, w} {f, b} 400 {b, c, k, s, p} {c, b, p} 500 {a, f, c, e, l, p, m, n} {f, c, a, m, p} *min_support = 3* 1. Scan DB once, find frequent 1-itemset (single item pattern) 2. Sort frequent items in frequency descending order, f-list 3. Scan DB again, construct FP-tree **Header Table** *Item frequency head* f 4 c 4 a 3 b 3 m 3 p 3 F-list = f-c-a-b-m-p  38	2	L3

	Find Patterns Having P From P-conditional Database ▪ Starting at the frequent item header table in the FP-tree ▪ Traverse the FP-tree by following the link of each frequent item *p* ▪ Accumulate all of *transformed prefix paths* of item *p* to form *p*'s conditional pattern base  40		
4	Explain the process of mining the colossal patterns with pattern fusion method with suitable example. Definition Colossal Frequent Patterns are very large frequent itemsets that contain a large number of items and appear frequently in a transaction database. Instead of generating all frequent itemsets (which can be exponentially many), Pattern Fusion directly constructs large (colossal) patterns by merging smaller frequent patterns. Pattern Fusion is an efficient approach because it avoids generating all subsets of a frequent itemset, thereby reducing computational complexity and memory usage. Need for Pattern Fusion Traditional algorithms like Apriori and FP-Growth generate a huge number of frequent itemsets before finding the largest ones. For example, Frequent Patterns A B C AB AC BC ABC ABCD ABCDE If the final objective is to obtain only ABCDE, generating all intermediate patterns is inefficient.	2	L3

	Pattern Fusion overcomes this problem by directly constructing larger patterns. Working Principle The algorithm starts with small frequent patterns. ↓ Find patterns having common items. ↓ Fuse (merge) them into larger patterns. ↓ Check support. ↓ If frequent, continue fusion. ↓ Stop when no further fusion is possible.		

SECTION B

Answer ANY ONE question **1 X 10 marks = 10 Marks**

<table>
<tr><th>Q. No</th><th>Questions</th><th>CO</th><th>RBT</th></tr>
<tr><td>5</td><td>The data in the table below represents the apparent temperature (ºF) vs. the relative humidity (%) in a room whose actual temperature is 72 °F. Use this data to solve the apparent temperature when the relative humidity reaches 110% using linear regression.
<table>
<tr><td>Relative Humidity(%)</td><td>0</td><td>10</td><td>20</td><td>30</td><td>40</td><td>50</td><td>60</td><td>70</td><td>80</td><td>90</td><td>100</td></tr>
<tr><td>Apparent Temperature (ºF)</td><td>64</td><td>65</td><td>67</td><td>68</td><td>70</td><td>71</td><td>72</td><td>73</td><td>74</td><td>75</td><td>76</td></tr>
</table>
Intercept(b) = (Σ y- a(Σ x))/ N
Slope = (N Σ xy-(Σ x)(Σ y))/ (N Σ x^2-(Σ x)^2)
Step 1: N= 11
Step 2: xy
<table>
<tr><th>x</th><th>y</th><th>xy</th><th>x^2</th></tr>
<tr><td>0</td><td>64</td><td>0</td><td>0</td></tr>
<tr><td>10</td><td>65</td><td>650</td><td>100</td></tr>
<tr><td>20</td><td>67</td><td>1340</td><td>400</td></tr>
<tr><td>30</td><td>68</td><td>2040</td><td>900</td></tr>
<tr><td>40</td><td>70</td><td>2800</td><td>1600</td></tr>
<tr><td>50</td><td>71</td><td>3550</td><td>2500</td></tr>
</table></td><td>3</td><td>L3</td></tr>
</table>

	60	72	4320	3600		
	70	73	5110	4900		
	80	74	5920	6400		
	90	75	6750	8100		
	100	76	7600	10000		
	550	775	40080	38500		

Slope= ((11 * 40080)– (550*775)) / ((11*38500)- (302500))
=0.116

Intercept = (775- 0.116*550)/11 = 64.65

Y= 64.65+0.116 * x

Y=64.65+0.116*110
=77.41

6

Explain the following with an example:

i. Entropy
ii. Information gain
iii. Gini index
iv. Tree Pruning

i. Entropy

Definition

Entropy is a measure of **impurity** or **uncertainty** in a dataset. It indicates how mixed the class labels are. It is commonly used in decision tree algorithms such as **ID3**.

Formula

$$Entropy(S) = -\sum_{i=1}^{n} p_i \log_2 p_i$$

where:

- p_i= Probability of class i

3 L3

Example

Consider a dataset of **10 students**.

Class	Count
Pass	6
Fail	4

Probability:

$$P(Pass) = \frac{6}{10} = 0.6$$
$$P(Fail) = \frac{4}{10} = 0.4$$

Entropy

$$= -(0.6\log_2 0.6 + 0.4\log_2 0.4)$$
$$= -(0.6 \times -0.737 + 0.4 \times -1.322)$$
$$= 0.971$$

Interpretation

- Entropy = 0 → Pure dataset (all belong to one class)
- Entropy = 1 → Maximum impurity (equal distribution)

ii. Information Gain

Definition

Information Gain (IG) measures the **reduction in entropy** after splitting the dataset based on an attribute. The attribute with the **highest information gain** is selected as the decision node in the ID3 algorithm.

Formula

$$IG(S,A) = Entropy(S) - \sum \frac{|\ S_v\ |}{|\ S\ |} Entropy(S_v)$$

where:

- S= Original dataset
- A= Attribute
- S_v= Subset after split

Example

Suppose:

Original dataset entropy = **0.971**

After splitting by **Attendance**:

Attendance	Pass	Fail	Entropy
Yes	5	1	0.650
No	1	3	0.811

Weighted entropy

$$= \frac{6}{10}(0.650) + \frac{4}{10}(0.811)$$
$$= 0.390 + 0.324$$
$$= 0.714$$

Information Gain

$$IG = 0.971 - 0.714$$
$$= 0.257$$

Interpretation

A higher information gain indicates a better attribute for splitting.

iii. Gini Index

Definition

The **Gini Index** is another measure of impurity used mainly in the **CART (Classification and Regression Tree)** algorithm.

A lower Gini Index indicates a better split.

Formula

$$Gini(S) = 1 - \sum p_i^2$$

Example

Using the same dataset:

Class	Probability
Pass	0.6
Fail	0.4

Gini Index

$$= 1 - (0.6)^2 - (0.4)^2$$
$$= 1 - 0.36 - 0.16$$
$$= 0.48$$

Interpretation

- Gini = 0 → Pure dataset
- Higher Gini → More impurity

iv. Tree Pruning

Definition

Tree Pruning is the process of **removing unnecessary branches** from a decision tree to reduce overfitting and improve the model's ability to generalize to unseen data.

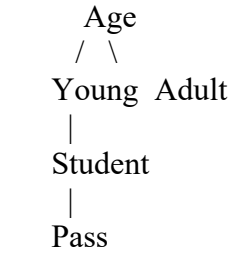
Types of Pruning

1. Pre-Pruning (Early Stopping)

The tree construction is stopped early based on conditions such as:

- Maximum tree depth
- Minimum number of samples per node
- Minimum information gain

Example



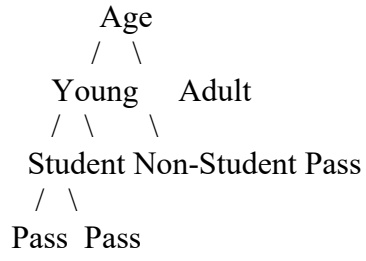
The algorithm stops expanding the tree once a stopping criterion is met.

2. Post-Pruning

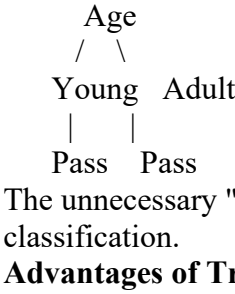
First, a complete tree is built. Then branches that do not improve prediction accuracy are removed.

Example

Before Pruning



After Pruning



The unnecessary "Student" split is removed because it does not improve classification.

Advantages of Tree Pruning

- Reduces overfitting
- Improves prediction accuracy
- Produces simpler and more interpretable trees

	• Reduces computation time		

Revised Bloom's Taxonomy (RBT) Levels:		
L1 – Remembering	L2 – Understanding	L3 – Applying
L4 – Analyzing	L5 – Evaluating	L6 – Creating